COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	**SECTION:**	CPE21S4
DATE PERFORMED:	10/01/26	**DATE SUBMITTED:**	10/01/26
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SEATWORK NO. 10.1 Graph

1. PROCEDURE OUTPUT

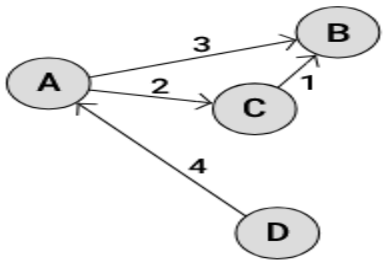
What procedures did you perform?

General Instructions:

1. Define the following: (10)

Graph
Vertex
Nodes
Weight
Path
Directed graph
Connected graph
Directed cyclic
Adjacency matrix
Adjacency List

2. Identify the parts of the following graph. (10)



3. Plot the adjacency matrix of the graph above. (10 pts)

4. Create a program in C++ program to create a graph that looks like in the above figure.

5. Differentiate BFS vs DFS types of search.

Answers:

1. Define the following:

A. Graph
- Is a data structure used to represent connections between different objects.

B. Vertex
- Is a point in a graph that represents an object or location.

C. Nodes
- Are the elements or points that make up a graph. They are connected by edges.

D. Weight
- Is a value given to an edge. It can represent things like distance, cost, or time.

E. Path
- Is a sequence of vertices connected by edges from one vertex to another.

F. Directed Graph
- Is a graph where the edges have a specific direction.

G. Connected Graph
- Is a graph where the vertices are connected so that there is a way to reach one vertex from another.

H. Directed Cyclic
- Is a directed graph that contains a cycle where you can follow the arrows and eventually return to the starting vertex.

I. Adjacency Matrix
- Is a table that shows the connections between vertices in a graph.

J. Adjacency List
- Is a list that shows the vertices connected to each vertex in a graph.

2. Identify the parts of the following graph.

Vertices: A, B, C, D
Edges: A -> B, A -> C, C -> B, D -> A
Total Vertices: 4
Total Edges: 4
Weights: 3, 2, 1, 4
Paths: A -> C -> B, D -> A -> B, D -> A -> C -> B

3. Plot the adjacency matrix of the graph above. (10 pts)

	A	B	C	D
A	0	3	2	0
B	0	0	0	0
C	0	1	0	0
D	4	0	0	0

|0 3 2 0|
|0 0 0 0|
|0 1 0 0|
|4 0 0 0|

4. Create a program in C++ program to create a graph that looks like in the above figure.

C/C++

```
#include <iostream>

int main() {
    char vertices[] = {'A', 'B', 'C', 'D'};

    int graph[4][4] = {
        {0, 3, 2, 0},  // A -> B = 3, A -> C = 2
        {0, 0, 0, 0},  // B
        {0, 1, 0, 0},  // C -> B = 1
        {4, 0, 0, 0}   // D -> A = 4
    };

    std::cout << "Graph:\n";

    std::cout << "    ";
    for (int i = 0; i < 4; i++) {
        std::cout << vertices[i] << "    ";
    }
    std::cout << std::endl;

    for (int i = 0; i < 4; i++) {
        std::cout << vertices[i] << "   ";
```

```
        for (int j = 0; j < 4; j++) {
            std::cout << graph[i][j] << "    ";
        }

        std::cout << std::endl;
    }

    return 0;
}
```

5. Differentiate BFS vs DFS types of search.
Based on my understanding Sir I think BFS searches the graph one level at a time but DFS searches by going deep into one path before backtracking.

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2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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